

Assignment 3

Vedika Yadav

Task 1 –

I have chosen a diversified basket of stocks as instructed from NSE, BSE and NASDAQ.

I first tried to import the entire list of stocks listed on these exchanges and then calculating the best of them based on Sharpe Ratio but I wasn't able to do this because I was running into time out error. So I chose a list of stocks based on the top 10 stocks of each of these exchanges over 5 years as given on Money Control and Screener.

The list of stock tickers I have used is as follows

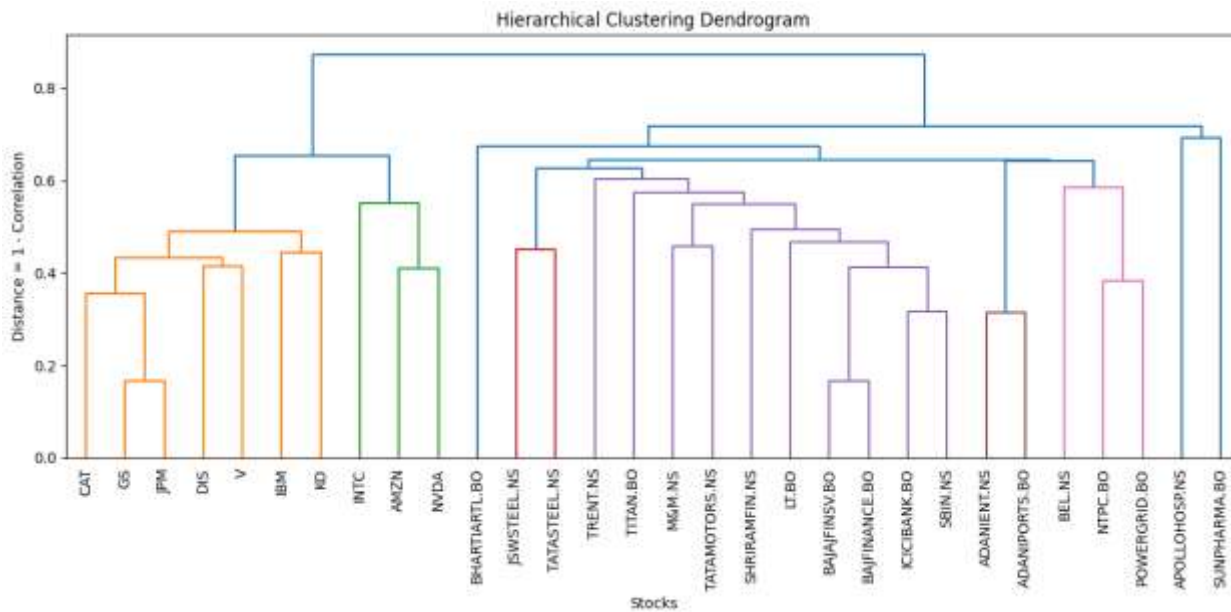
```
tickers = ['ADANIENT.NS', 'BEL.NS', 'TRENT.NS', 'TATAMOTORS.NS', 'M&M.NS', 'SHRIRAMFIN.NS',  
'JSWSTEEL.NS', 'APOLLOHOSP.NS', 'TATASTEEL.NS', 'SBIN.NS', 'ADANIPORTS.BO',  
'ICICIBANK.BO', 'BAJFINANCE.BO', 'LT.BO', 'BAJAJFINSV.BO', 'TITAN.BO', 'SUNPHARMA.BO',  
'NTPC.BO', 'BHARTIARTL.BO', 'POWERGRID.BO', 'NVDA', 'JPM', 'GS', 'DIS', 'AMZN', 'INTC', 'V', 'IBM',  
'CAT', 'KO']
```

Task 2 –

I downloaded the data for all the tickers from yfinance and then used the adjusted close values to calculate daily log returns and correlation matrix.

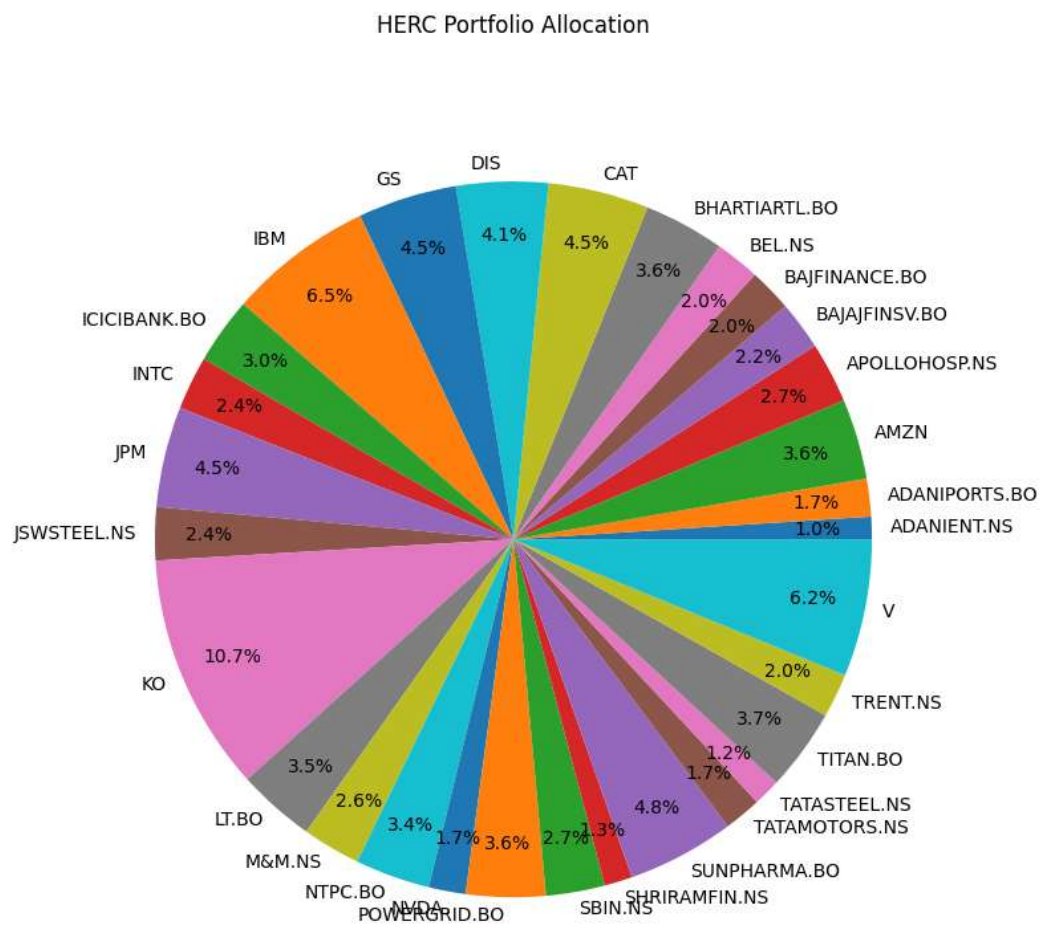
Task 3 –

For making the hierarchical clusters I used linkage library from scipy.cluster.hierarchy and use the dendrogram library to plot the dendrogram as instructed. I first converted the correlation_matrix to a distance_matrix because the more the correlation between two stocks the lesser will be there distance in the clustering process. Then we converted the matrix to a condensed format since that's needed for us to use linkage. The dendrogram produced is



Task 4 –

Since HERC builds on HRP and is better at weight allocation I decided to use that. First I used the riskfolio-lib and for a rolling window of 126 days that translates to about 6 months I calculated the weights using a linear codependence model ie. Pearson and using ward because it is preferred over others. I also plotted a pie chart showing the asset allocation



Task 5 –

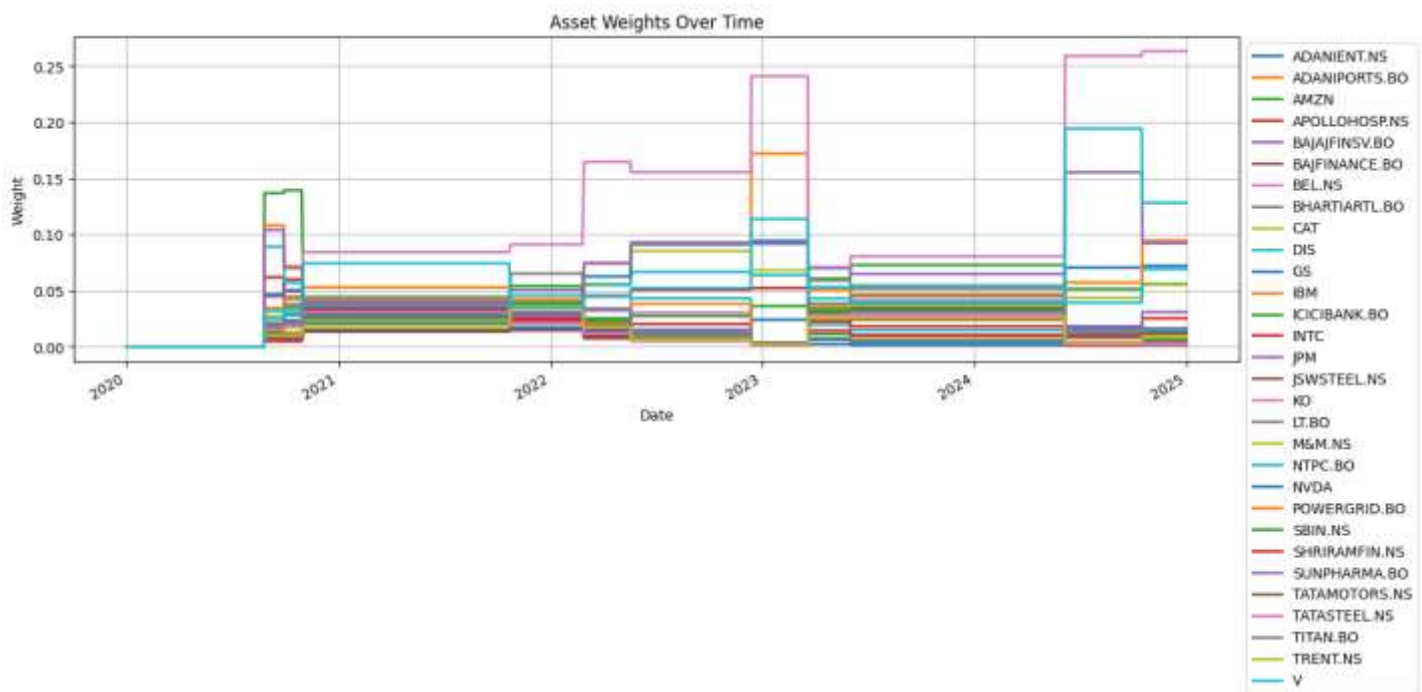
We had been instructed to make a model that would rebalance when necessary. So, now taking the same lookback period but using a no-rebalance time of 21 days along with it, I chose a threshold of 5% for the correlation. Calculated the correlation matrix again and set up the conditions for rebalancing.

Task 6 –

When we plot the graph for with cost portfolio returns I have added the transaction cost. I haven't added it earlier than that because we need to compare two separate portfolios, one with and one without cost. For the without cost one we have to ensure that we don't deduct the transaction cost. For the with cost one we ensure that the money invested after rebalancing is less than what we had withdrawn.

Task 7 –

Final Asset Allocation –



Portfolio Performance vs Benchmark and Portfolio With and Without Costs

